Laboratory Report

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Student Declaration

1. I have accurately identified and included the sources of all facts, ideas, opinions, and viewpoints from others in the assignment references. All direct quotations, paraphrasing, and discussions of ideas from books, journal articles, internet sources, course materials, or any other sources used are properly acknowledged and cited in the assignment references.
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1. Description

This document lists 11 batch-file demonstrations (from a Week 5 Infrastructure as Code document) as well as an exercise for a backup-script. Batch-file structure and directory listing, pausing the batch, variables and their usage, command-line arguments, user input, checking the existence of files, error handling, output redirection, and file/directory looping were demonstrated. The practical was conducted on a virtual machine of Windows 11 x64 operating system installed in VMware Workstation Pro 26H1. Scripts were edited using Notepad++ and then executed by using the windows command prompt and managed practical repository and evidence using Git with GitHub Desktop.

Five sample text files were used for the backup exercise and a batch file, Backup.bat, was created to perform the following: Create a directory with the current date, Copy the five sample text files, and Set the Read Only attribute to the copied files. On first test, handover records that no files are being copied due to an invalid path. The following test showed “5 File(s) copied”. The output of the attrib command listed the R attribute for all 5 of the files and the File Explorer listed the 5 files in a dated folder. The date of the observed folder was 26OCT26 but the assignment said 25OCT26. Overall the results show that scripting is useful for repeatable file-management tasks and, that there is need for error checking, verification and compliance with the requirements as specified.

2. Aims

This practical aimed to develop and evaluate Windows batch scripting through 11 DOS demonstrations and a backup exercise. It also examined the importance of testing, verification and error handling.

The specific aims were:

1. To complete and test all 11 DOS batch-file demonstrations provided in the Week 5 materials.
2. To organise scripts, sample data, backups and test evidence in the IAC repository.
3. To develop and test Backup.bat for creating a dated directory using the DDMMMYY format.
4. To copy five sample text files and verify their presence in the backup directory.
5. To apply and verify the read-only attribute using the attrib command.

6. To evaluate the results, identify limitations and recommend evidence-based improvements.

3. Equipment

Item	Description	Make/Model	Details
1	Virtualisation software	VMware Workstation Pro 26H1	N/A
2	Virtual machine	Windows 11 x64	Win 11 Dev
3	Text editor	Notepad++	Version not specified
4	Version control software	Git and GitHub Desktop	Versions not specified

3.1 Practical Environment

The practical was performed in a virtual machine (VMware Workstation Pro 26H1) with the Windows 11 x64 operating system. The working machine (Win 11 Dev) was created as a complete copy of the virtual machine Windows 11 x64 GOLD. It was set up with “4 GB of memory”, “2 processors”, a “64 GB virtual disk” and a “NAT network adapter”. GOLD virtual machine was not turned on as a part of the exercises. This was the setup in which the batch scripts have been developed and run. Testing was done using the Windows command prompt to execute the scripts and to view the results.

3.2 Software and Repository

The Batch files were created and edited using Notepad++ and run and tested using the Windows command prompt. The repository was managed by Git and GitHub Desktop. The IAC repository comprised four main directories – DOS/BatchFiles/ held the files required to execute the scripts, DOS/Data/ held the five sample text files, DOS/Backups/ held the dated backup folders, and DOS/Tests/ held the screenshots of the demonstration outputs and testing results. This allowed for separation of scripts, source data, generated backups and evidence; helping to easily locate and review the practical files.

4. Method

The practical included setting up a Windows 11 virtual machine and doing 11 demos of DOS batch files, as well as writing a back-up script. Scripts were edited in Notepad++ and then run and results checked in the Windows command prompt.

4.1 Environment Setup

The Win11 Dev virtual machine was set-up as a complete cloned machine from the Win11 x64 GOLD virtual machine in VMware Workstation Pro 26H1. The working environment consisted of 4 GB of memory, 2 processors, 64 GB virtual disk and NAT network adapter. IAC repository was locally cloned using GitHub Desktop. The DOS directory was split into BatchFiles, Data, Backups and Tests to keep scripts, original files, generated backups and screenshots in their own directories.

4.2 DOS Batch-File Demonstrations

All of the eleven batch-file demonstrations were run to explore various batch-file techniques in Windows. Each script was checked by viewing its output in the command window. Table 1 summarises the purpose and procedure of each demonstration.

Filename	**Purpose and procedure**
Demo1.bat	Demonstrates directory listing by using the dir /w command to display the contents of the current directory in a wide format.
Demo2.bat	Demonstrates basic batch-file structure by displaying a title, header and introductory banner using echo commands.
Demo3.bat	Demonstrates a pause before a critical operation. The script displays a warning and waits for user input before continuing.
Demo4.bat	Demonstrates variable handling using SETLOCAL and ENDLOCAL. It displays the virtual-machine path and performs an arithmetic calculation.
Demo5.bat	Demonstrates command-line arguments by accepting two supplied values and displaying them in a welcome message. The script was tested with pawan venugopal.
Demo6.bat	Demonstrates interactive user input using set /p. The user enters a name, which the script then displays.
Demo7.bat	Demonstrates conditional file-existence checking using an if exist statement to check whether githubdesktop.exe exists in the Windows directory.
Demo8.bat	Demonstrates error-level checking by attempting to copy DoesNotExist.exe and using IF %ERRORLEVEL% NEQ 0 to identify an unsuccessful operation.
Demo9.bat	Demonstrates success and failure handling using && and `

Demo10.bat	Demonstrates output redirection by creating SimpleBackup.log and appending copy-operation output and error messages to the log file.
Demo11.bat	Demonstrates file and directory iteration using FOR and FOR /D loops to display filenames in the current directory and directories within the user's profile.

The corresponding script listings are provided in Appendix A. The observed outputs and test results are discussed in Section 5.

4.3 Backup Script Development

The requirements for writing the backup.bat program were to display explanatory messages, get and format the date, make a directory to store backups under DOS/Backups, copy five sample text files from DOS/Data, set the read-only attribute on the copies, and print a completion message. The observed destination was 26OCT26 which is in DDMMMYY format. The difference will, however, be noted, but not silently altered; it is specified in the assignment: '25OCT26'. The source code for Backup.bat is listed in Appendix B.

4.4 Testing Procedure

The backup script was run in the command-interactive mode. At the first try, it reported an invalid path and 0 file were copied. After checking the paths the subsequent run indicated that 5 File(s) were copied. Afterwards, they examined the destination and its five files, using File Explorer and attributing the R attribute using the attrib tool. The complete code correction was not recorded during handover.

5. Results and Testing

The results were evaluated using the supplied screenshots, command outputs and practical handover. Testing examined the behaviour of the 11 DOS demonstrations, successful file copying, destination contents and file attributes.

5.1 Results of the DOS Demonstrations

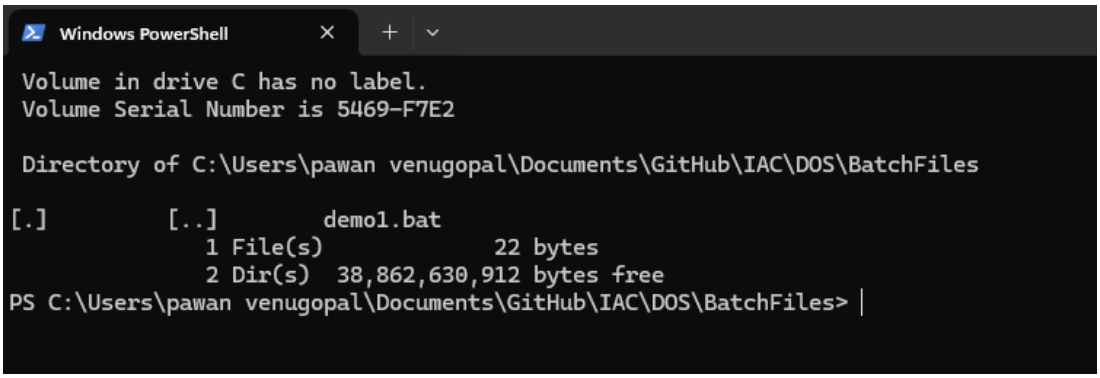


Figure 1: Directory listing

Observed result: The screenshot shows the BatchFiles directory, including demo1.bat and the current and parent directory entries.

Evaluation: This shows that the file was in the working directory, but the screen shot by itself does not show what the result was when the Demo1.bat file was ran.

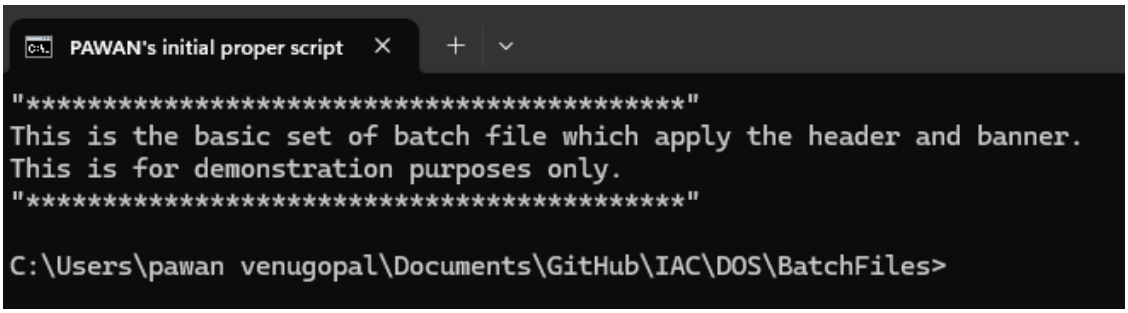


Figure 2: Batch-file header

Observed result: The script showed its welcome banner.

Evaluation: This helps address the behaviour of displaying the expected message.

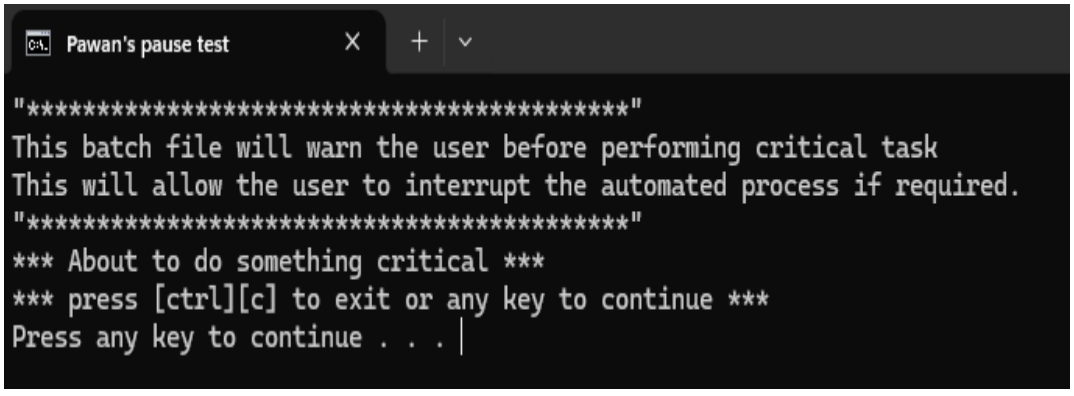


Figure 3: Pause test

Actual result: The script did not show a warning and did not wait for a keypress input.

Evaluation: This is the total score for the pause prompt.

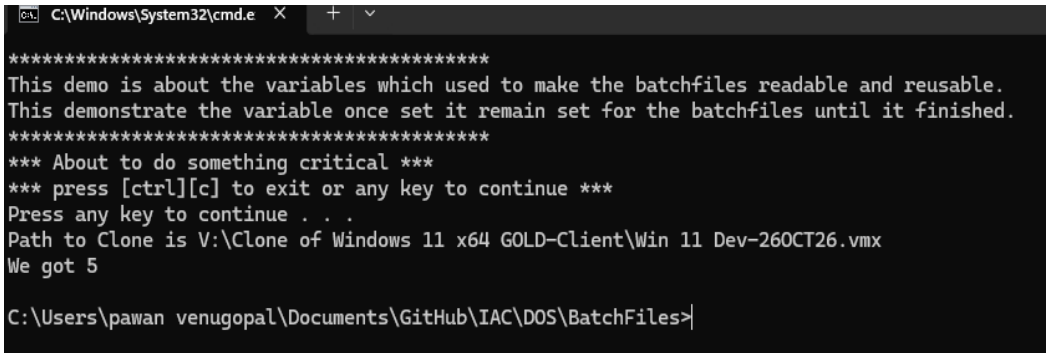


Figure 4: Variable test

Observed result: Virtual machine path and value 5 was displayed.

Evaluation: This shows variable uses.

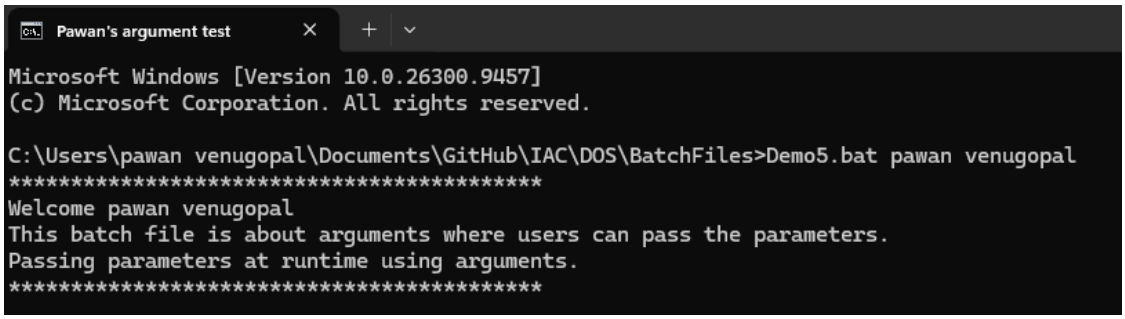


Figure 5: Argument test

Observed result: The command displayed Welcome pawan venugopal.

Evaluation: This helps with the use of command-line arguments.

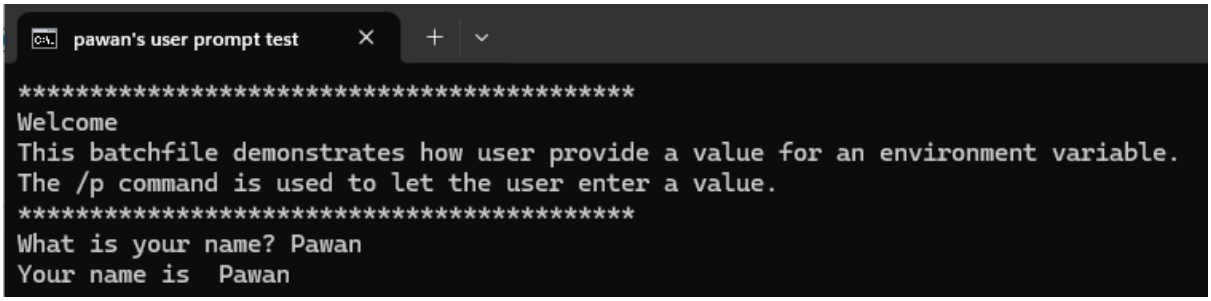


Figure 6: User prompt

Observed result: Pawan had been accepted by the script and the entered name was displayed.

Evaluation: This check for interactive input of the value to be tested.

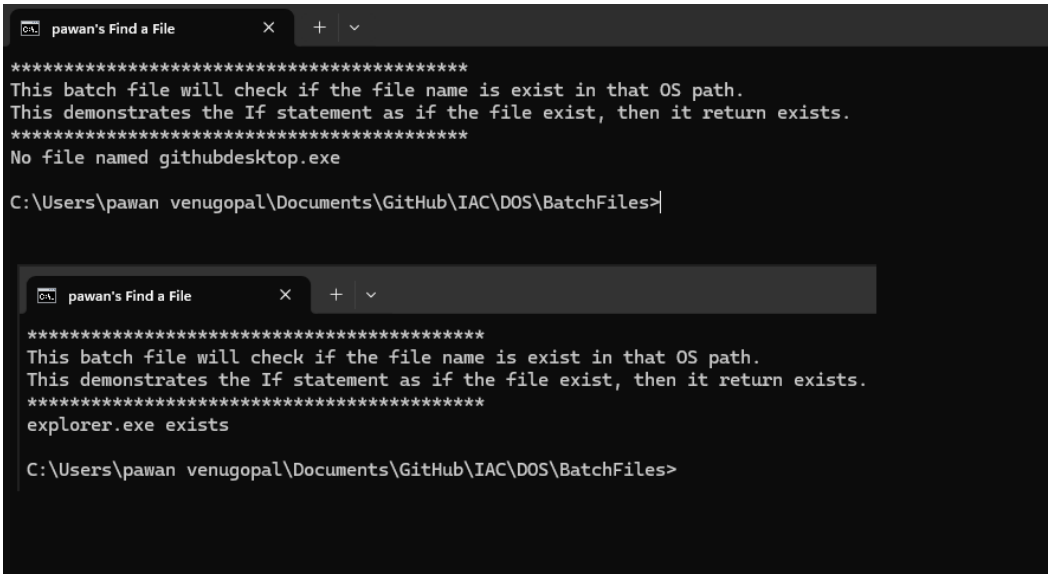


Figure 7: File-existence test

Observed result: The outputs captured indicate that in one case, githubdesktop.exe was not found, and in the other, explorer.exe was reported as existing.

Evaluation: These results show the file-existence checking for the tested file names. The outputs should be interpreted as individual tests and not that the same unaltered script tested both files with a single run.

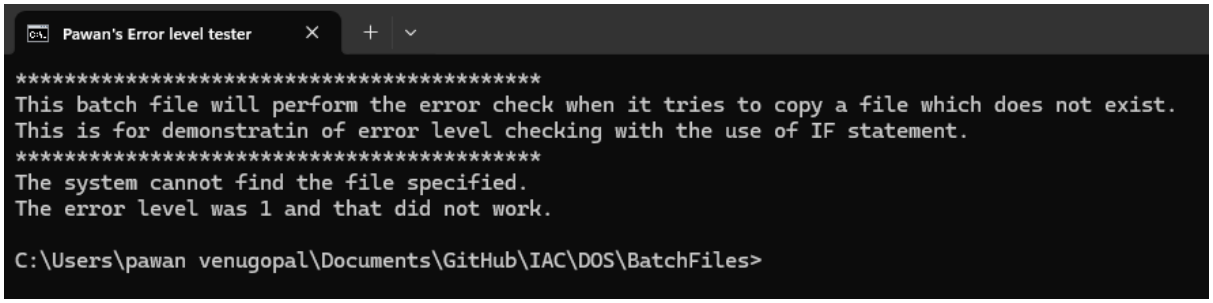


Figure 8: Error-level test

Observed result: The copy attempt for the missing file resulted in an error level of 1.

Evaluation: This is used to help detect failure being tested.

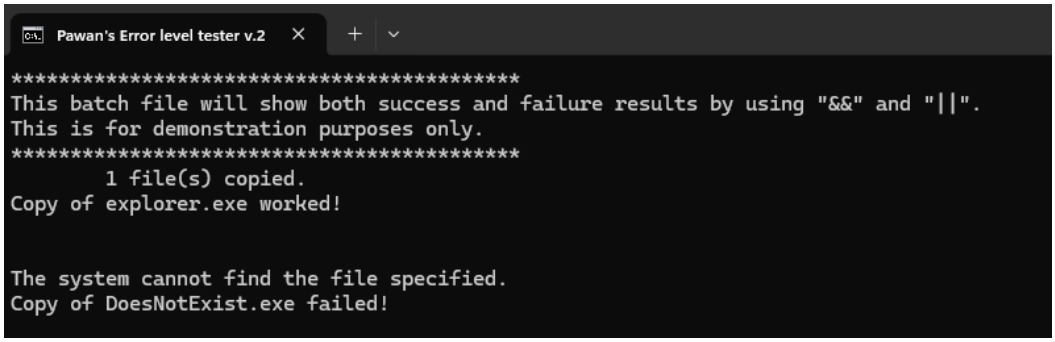


Figure 9: Conditional copy test

Observed result: Copying explorer.exe succeeded, DoesNotExist.exe failed. **Evaluation:** output has succeeded/fail handling with && and ||.

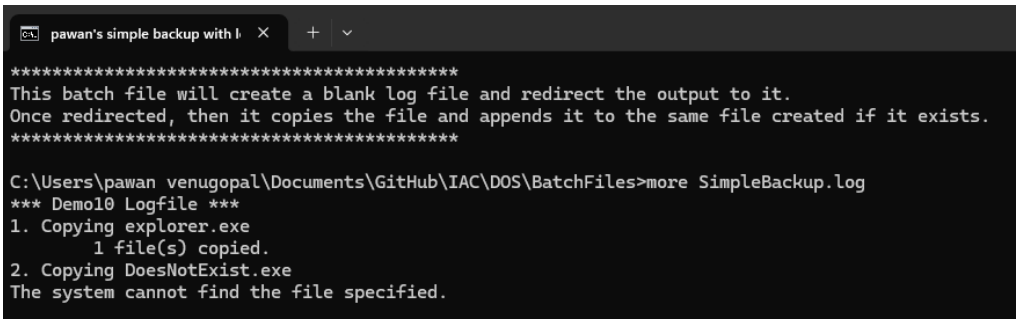


Figure 10: Output logging

Observed result: Both copy results were logged in SimpleBackup.log. **Evaluation:** This demonstrates output redirection for subsequent inspection.

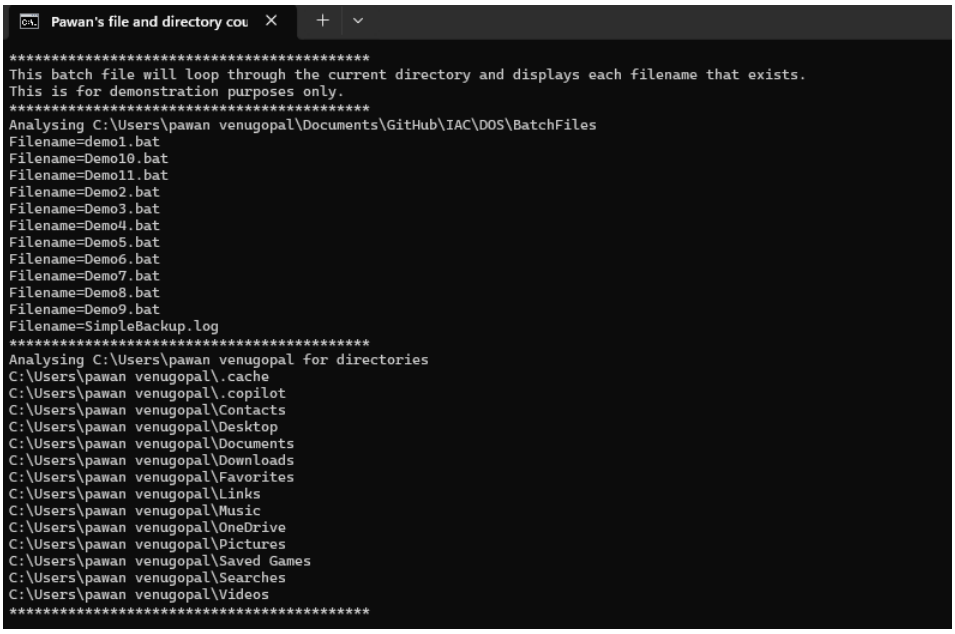


Figure 11: File and directory loops

Observed result: The script listed files and profile directories.

Evaluation: This is used with FOR and FOR /D loops.

5.2 Backup Script Results

The initial backup reported that its path was invalid and it could copy 0 files. After checking out the path, the next test indicated 5 File(s) copied, as per the practical handover.

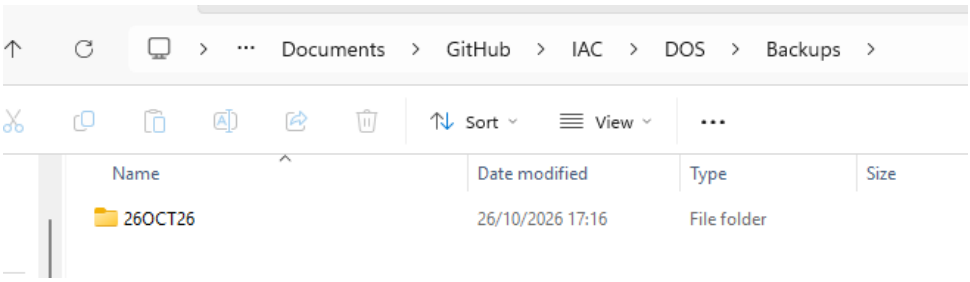


Figure 12: Dated backup directory

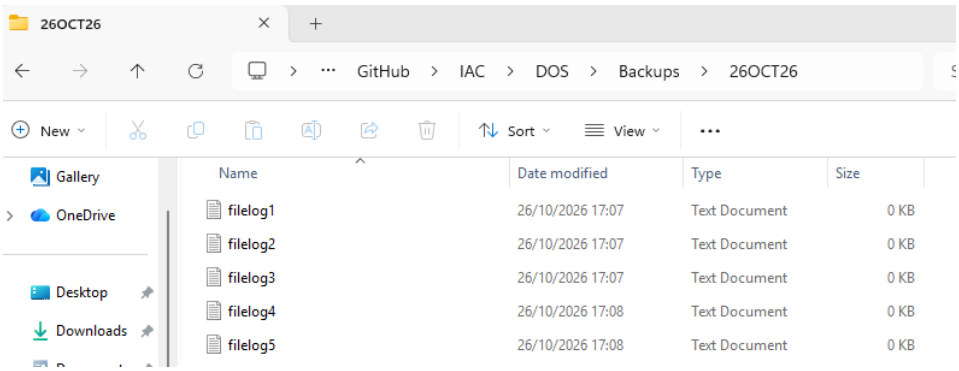


Figure 13: Five copied files

The screenshots depict the 26OCT26 folder, which contains five text files. This helps to successfully duplicate the sample data set.

5.3 File Attribute Verification

```
C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS>attrib "Backups\26OCT26\*.*"
A    R              C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS\Backups\26OCT26\filelog1.txt
A    R              C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS\Backups\26OCT26\filelog2.txt
A    R              C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS\Backups\26OCT26\filelog3.txt
A    R              C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS\Backups\26OCT26\filelog4.txt
A    R              C:\Users\pawan venugopal\Documents\GitHub\IAC\DOS\Backups\26OCT26\filelog5.txt
```

Figure 14: Read-only attribute test

All five files had A and R in the attrib output. The R attribute can aid in verification of read only, but cannot guarantee full backup protection or recovery.

5.4 Summary of Test Results

Test	Expected outcome	Actual result	Evaluation
DOS demonstrations	Intended behaviour shown	Outputs visible in screenshots	Supports tested behaviours
Initial backup	Files copied	Invalid path; zero copied	Failed
Subsequent backup	Five files copied	5 File(s) copied	Successful for tested files
Destination inspection	Five files present	Five files visible	Supports copying
Attribute verification	Read-only applied	R displayed	Supports attribute verification
Backup date	25OCT26, as specified in the assignment instructions	26OCT26 observed	The recorded result does not match the specified date. Clarification is required.

In general the most successful results include a successful “copy” and files that are read-only and that you can see the destination files. But it is seen that the observed 26OCT26 is not the same as the requirement of the assignment which is 25OCT26. There was no evidence found that a back-up was used to fix the issue after the initial failure and what the exact script was to correct the issue.

6. Conclusions

6.1 Achievement of Aims

The main objectives of the practical were fulfilled. 11 DOS demonstrations were given on variables, user input, arguments, conditional statements, error handling, logging and loops. The backup test then reported 5 File(s) copied, and the screen shots indicated 5 files in the destination directory all with the R attribute set for each file. However, there is a date requirement issue since the folder observed was 26OCT26, while the assignment is for 25OCT26. Also, restoration of the backup was not tested from the available evidence.

6.2 Technical Significance

The demonstrations illustrated how batch scripts can be used to string together commands to make repeatable procedures for file-management. Variables prevent data from being re-

entered, conditional statements enable decision making and logging provides a record for trouble shooting. Doing the first backup fails is proof that execution doesn't do execution. For the five sample files, checking command outputs and looking at the location after the move provided better evidence that the move itself had been successful.

6.3 Independent Research

Microsoft documents xcopy exit codes that could be tested in scripts to check if copying was successful or to check for certain failure conditions [1]. This affects the first "failed-path" and the last test (that was successful). More reliable script should check the return of the command before reporting the completion. This was necessary and demonstrated with the practical, but it was not proven that all of the possible scenarios for error were considered with the evidence.

Microsoft states that the attrib command works to display and change file attributes like read-only and archive attributes [2]. The R attribute observed indicates that the copies files had the read only attribute. It cannot be assumed, however, that the contents are the same as the originals. Comparing the sources with the destinations would be more a proof of integrity of data and will give more than attribute check testing.

The National Institute of Standards and Technology advises for items such as backups to be made regularly, tested or reviewed through recovery exercises [3]. While its guidance is with regards to Operational Technology (OT), the principles apply when thinking about backup reliability as a whole. It was confirmed during current practical that 5 files were copied but there was no record in the log of a restoration test. Thus, the evidence does not show successful recovery in the tested scenario, but successful copying. More security would result from a restoration test.

The Pro Git book documents that commits are a snapshot of the changes within a repository which can be reviewed and compared [4]. A combination of consistent filenames and clear figure captions can enhance the traceability of the batch scripts and screenshots in the IAC repository. This would be useful to link the result to the corresponding script version. However, version control doesn't replace a backup/recovery procedure.

6.4 Limitations and Improvements

In the exercise, only 5 sample files were used, so there is not as much evidence about the performance of using larger files or with more complex directory structures. The first time, the procedure failed, indicating it needed trouble shooting to be successful. Future

enhancements shall involve validation of source and target paths, verification of command exit codes, verification of number of files copied, verification of content of file(s) copied. Also include recovery testing. Lastly, the date difference must be resolved with the lecturer/official instructions, and the result must not be altered.

6.5 Overall Conclusion

Overall the practical was useful in terms of batch-scripting techniques and successful evidence of copying and verification of the read only attribute. The date issues, the small test set used, and the lack of a restoration test, however, limit what can be concluded with regards to full compliance and backup reliability. The validation of the procedure with additional validation/failure-condition testing and recovery checks would make the procedure more reliable and provide greater evidence of its suitability for multiple reusages.

References

[1] Microsoft, “xcopy,” *Microsoft Learn*. [Online]. Available: https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/xcopy.

[2] Microsoft, “attrib,” *Microsoft Learn*. [Online]. Available: https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/attrib

[3] National Institute of Standards and Technology, *OT Backup Quick Start Guide*, NIST SP 1339, June 2026. [Online]. Available: https://csrc.nist.gov/pubs/sp/1339/final

[4] S. Chacon and B. Straub, “Recording Changes to the Repository,” *Pro Git*, 2nd ed. [Online]. Available: https://git-scm.com/book/en/v2/Git-Basics-Recording-Changes-to-the-Repository.html

Appendices

Appendix A: DOS Demonstration Scripts

A.1 Demo1.bat - Directory Listing

```
:: Demo batch file to show the directories
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo1.bat

@echo off
cls
dir /w
```

A.2 Demo2.bat - Batch-File Header and Banner

```
:: Demo batch file to show a header and banner
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo2.bat

@echo off
cls
title PAWAN's initial proper script writing
echo "*******************************************"
echo This is the basic set of batch file which apply the header and banner.
echo This is for demonstration purposes only.
echo "*******************************************"
```

A.3 Demo3.bat - Pause and User Interruption

```
:: Demo batch file to allow user interrupt
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCTC26
:: Filename: Demo3.bat

@echo off
cls
title Pawan's pause test
echo "*******************************************"
echo This batch file will warn the user before performing critical task
echo This will allow the user to interrupt the automated process if required.
echo "*******************************************"
echo *** About to do something critical ***
echo *** press [ctrl][c] to exit or any key to continue ***
pause
```

A.4 Demo4.bat - Variables

```
:: Batch file to demonstrate variables
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo4.bat

@echo off
cls
title Pawan's variable test
echo ****************************************
echo This demo is about the variables which used to make the batchfiles readable and reusable.
echo This demonstrate the variable once set it remain set for the batchfiles until it finished.
echo ****************************************

echo *** About to do something critical ***
echo *** press [ctrl][c] to exit or any key to continue ***
pause

SETLOCAL

SET clonepath=V:\Clone of Windows 11 x64 GOLD-Client\
SET clonename=Win 11 Dev-26OCT26.vmx
echo Path to Clone is %clonepath%%clonename%

SET /A calculation=2+12/4
echo We got %calculation%

ENDLOCAL

EXIT
```

A.5 Demo5.bat - Command-Line Arguments

```
:: Batch file to demonstrate arguments
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo5.bat

@echo off
:: cls
Title Pawan's argument test
echo ********************************************
echo Welcome %1 %2
echo This batch file is about arguments where users can pass the parameters.
echo Passing parameters at runtime using arguments.
echo ********************************************
```

A.6 Demo6.bat - User Input

```
:: Batch file to take arguments at runtime
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo6.bat

@echo off
cls
title pawan's user prompt test
echo ******************************************
echo Welcome
echo This batchfile demonstrates how user provide a value for an environment variable.
echo The /p command is used to let the user enter a value.
echo ******************************************

set /p NAME=What is your name?
echo Your name is %NAME%
```

A.7 Demo7.bat - File-Existence Check

```
:: Batch file to demonstrate conditional branching
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26JAN26
:: Filename: Demo7.bat

@echo off
cls
title pawan's Find a File
echo ******************************************
echo This batch file will check if the file name is exist in that OS path.
echo This demonstrates the If statement as if the file exist, then it return exists.
echo ******************************************

SET ospath=C:\Windows\
SET filename=githubdesktop.exe

If exist %ospath%%filename% (
    echo %filename% exists
) else (
    echo No file named %filename%
)
```

A.8 Demo8.bat - Error-Level Checking

```
:: Demo batch file to check the error by using operators
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo8.bat

@echo off
cls
title Pawan's Error level tester
echo ********************************************
echo This batch file will perform the error check when it tries to copy a file which does not exist.
echo This is for demonstratin of error level checking with the use of IF statement.
echo ********************************************

SET ospath=C:\Windows\
SET filename=DoesNotExist.exe

:: The temp variable is defined globally for every user
Copy %ospath%%filename% %TEMP%

IF %ERRORLEVEL% NEQ 0 (
 echo The error level was %ERRORLEVEL% and that did not work.
)
```

A.9 Demo9.bat - Success and Failure Handling

```
:: Batch file to display success or failure
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26JAN26
:: Filename: Demo9.bat

@echo off
cls
title Pawan's Error level tester v.2
echo ********************************************
echo This batch file will show both success and failure results by using "&&" and "||".
echo This is for demonstration purposes only.
echo ********************************************

SET ospath=C:\Windows\
SET filename1=explorer.exe
SET filename2=DoesNotExist.exe
copy %ospath%%filename1% %TEMP% && echo Copy of %filename1% worked!
echo.
echo.
copy %ospath%%filename2% %TEMP% || echo Copy of %filename2% failed!
```

A.10 Demo10.bat - Output Redirection and Logging

```
:: Batch file to demonstrate redirection
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26JAN26
:: Filename: Demo10.bat

@echo off
cls
title pawan's simple backup with logging
echo ********************************************
echo This batch file will create a blank log file and redirect the output to it.
echo Once redirected, then it copies the file and appends it to the same file created if it exists.
echo ********************************************

:: Create a blank file or erase it if it already exists
echo *** Demo10 Logfile *** > SimpleBackup.log

SET ospath=C:\Windows\
SET filename1=explorer.exe
SET filename2=DoesNotExist.exe

echo 1. Copying %filename1% >> SimpleBackup.log
copy %ospath%%filename1% %TEMP% >> SimpleBackup.log 2>&1
echo 2. Copying %filename2% >> SimpleBackup.log
copy %ospath%%filename2% %TEMP% >> SimpleBackup.log 2>&1
```

A.11 Demo11.bat - File and Directory Loops

```
:: Batch file to demonstrate loops
:: By: PAWAN
:: Based on initial course documented by JOR
:: Initial file: 26OCT26
:: Filename: Demo11.bat

@echo off
cls
title Pawan's file and directory counter
echo ********************************************
echo This batch file will loop through the current directory and displays each filename that exists.
echo This is for demonstration purposes only.
echo ********************************************

:: Iterate through all the files in the current directory
echo Analysing %CD%
FOR %%I IN (*) DO @ECHO Filename=%%I
echo ********************************************
:: The environment variable USERPROFILE points at the users home folder
echo Analysing %USERPROFILE% for directories
FOR /D %%I IN ("%USERPROFILE%\*") DO @ECHO %%I
echo ********************************************
```

Appendix B: Backup Script

B.1 Backup.bat - Source Code for the Dated Backup Process

```
:: Batch script to copy all the files from Data folder to Backups folder by creating the folder with DDMONYY
:: By: PAWAN
:: Initial file: 26OCT26
:: Filename: Backup.bat

@echo off
cls
title Pawan's Backup Script

echo "************************************************"
echo Creating a dated backup folder inside the Backups by coping files from Data folder.
echo "************************************************"

:: Setting up the date components
set "DD=%DATE:~0,2%"
set "MM=%DATE:~3,2%"
set "YYYY=%DATE:~8,2%"

:: Convert month number to three letter month name using if statement
if "%MM%"=="01" set "MON=JAN"
if "%MM%"=="02" set "MON=FEB"
if "%MM%"=="03" set "MON=MAR"
if "%MM%"=="04" set "MON=APR"
if "%MM%"=="05" set "MON=MAY"
if "%MM%"=="06" set "MON=JUN"
if "%MM%"=="07" set "MON=JUL"
if "%MM%"=="08" set "MON=AUG"
if "%MM%"=="09" set "MON=SEP"
if "%MM%"=="10" set "MON=OCT"
if "%MM%"=="11" set "MON=NOV"
if "%MM%"=="12" set "MON=DEC"

set "BackupDate=%DD%%MON%%YYYY%"

:: Create the backup folder
if not exist "Backups\%BackupDate%" mkdir "Backups\%BackupDate%"

:: Copy the files inside data folder to Backups folder with correct dated format
xcopy "Data\*.*" "Backups\%BackupDate%\" /I/Y

:: Mark the backup files as read only
attrib +R "Backups\%BackupDate%\*.*" /S

echo.
echo Backup completed: %BackupDate%
pause
```